

Battery SOC Benchmark Platform

batterysocbenchmark.ca

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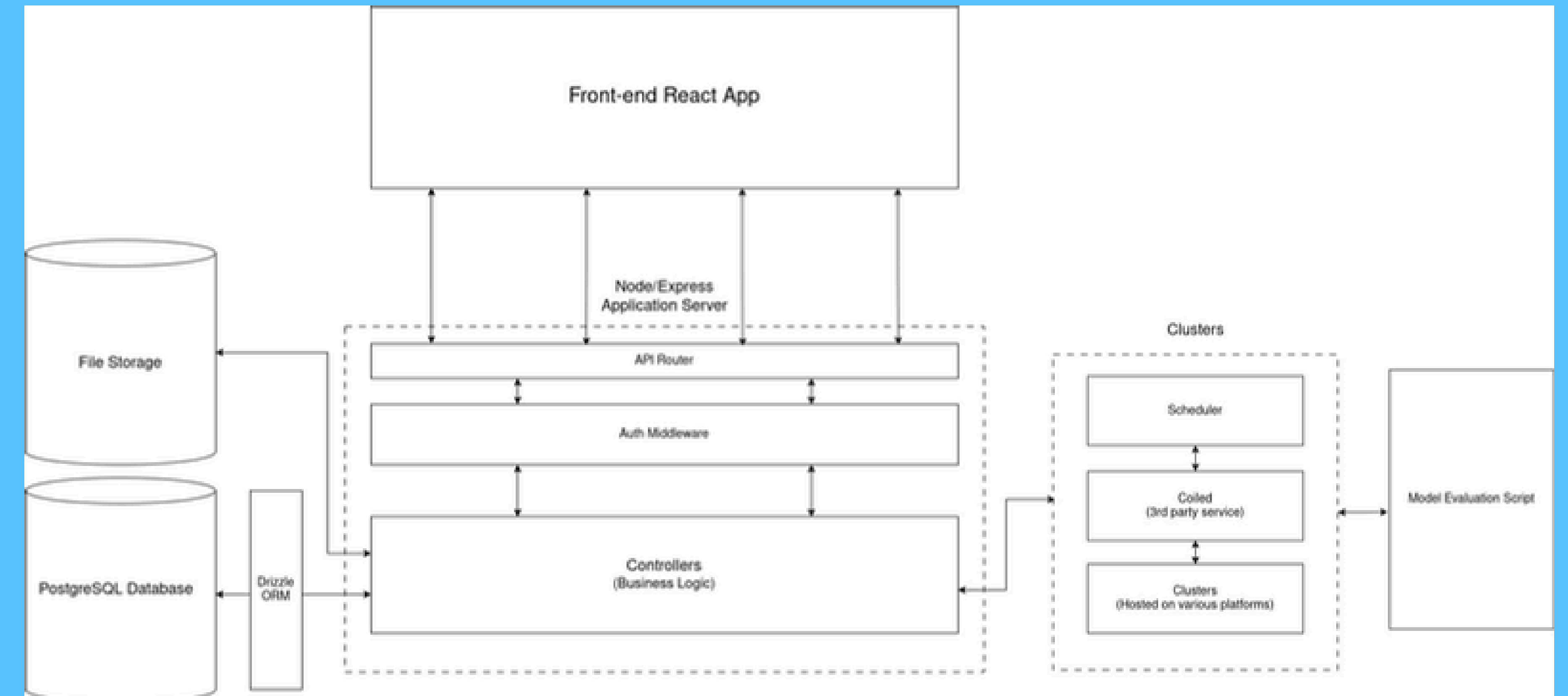
Motivations

Battery state of charge (SOC) estimation requires specialized algorithms and standardized testing to determine which SOC estimation approach is optimal. A centralized platform to crowd-source SOC algorithms and efficiently evaluate them will accelerate research to better develop battery products, benefiting researchers, engineers, and hobbyists alike.

We aim to help Dr. Phil Kollmeyer, an assistant professor at McMaster University in Electrical & Computing Engineering, achieve this.

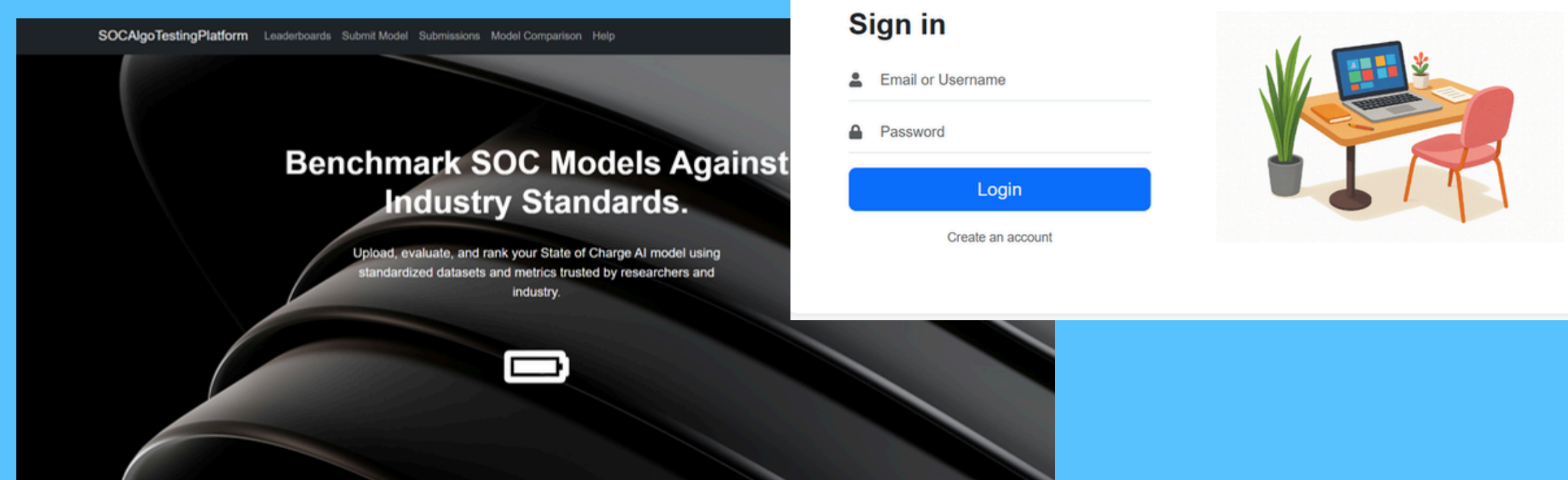
System Architecture

The system is a full-stack application, with a frontend React Client communicating with an application server through a RESTful API that handles authentication, validation, routing, and business logic. Compute-intensive model evaluations are offloaded asynchronously to a scheduling service using Coiled and Dask. Evaluation results and file metadata, as well as users, accounts, and sessions, are stored in a PostgreSQL database using Drizzle ORM. Authentication is handled by the Better-Auth library, and session cookies are used to protect endpoints, authorize users, and verify identity.

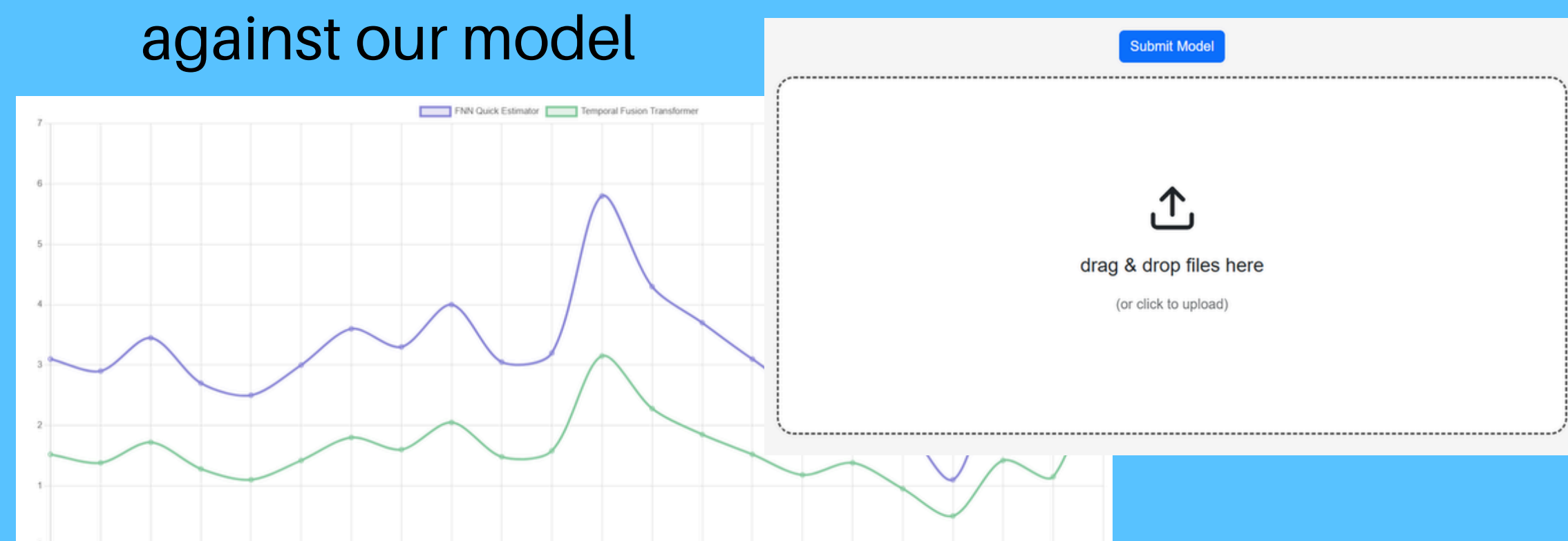


Platform Features

- Registration and Login



- Submit your Battery SOC models to benchmark against our model



- Track how your model compares to others on the leaderboards

Leaderboards										
Filter by Author			Filter by Academic Affiliation			Model Type				
All authors			All Academic Affiliations			All Model Types				
Submission	Model Name	Model Type	Status	Visibility	Submitted at	Completed at	Weighted Error	All Cells	Blind Cells	Non-Blind Cells
34	Physics-Informed Neural Hybrid	Hybrid Model	pending	Public	3/25/2026, 2:24:48 PM	3/25/2026, 2:24:48 PM	1.48	1.35	1.68	1.25
23	Temporal Fusion Transformer	Transformer	pending	Public	3/25/2026, 2:24:48 PM	3/25/2026, 2:24:48 PM	1.52	1.38	1.72	1.28
22	Transformer SOC Network	Transformer	pending	Public	3/25/2026, 2:24:48 PM	3/25/2026, 2:24:48 PM	1.65	1.5	1.88	1.4
17	Reinforcement LSTM SOC	LSTM	pending	Public	3/25/2026, 2:24:48 PM	3/25/2026, 2:24:48 PM	1.72	1.58	1.92	1.48
								1.3	1.65	2.05

- Chart multiple models to compare performance grades between them
- Comment on other user's model submissions to create communities

Parallelization

SOC evaluation runtimes can vary between a few minutes to several days. In order to improve responsiveness for a userbase of possibly hundreds of users doing back-to-back submissions, it is necessary to distribute work between several clusters of distributed cloud computers. Running in parallel, these worker computers follow the commands of a single scheduler, breaking up the data into smaller chunks that can be processed in parallel.

Parallelization Life Cycle:

- Client Layer** (scheduler host) submits task to Coiled
- Coiled** organizes clusters, orchestrates virtual machines from connected Cloud Providers
- Cloud Computing Platforms** execute tasks, returns result packets to Coiled, which packages them and returns the full, ordered solution to the **Client Layer**



Python Parallelization

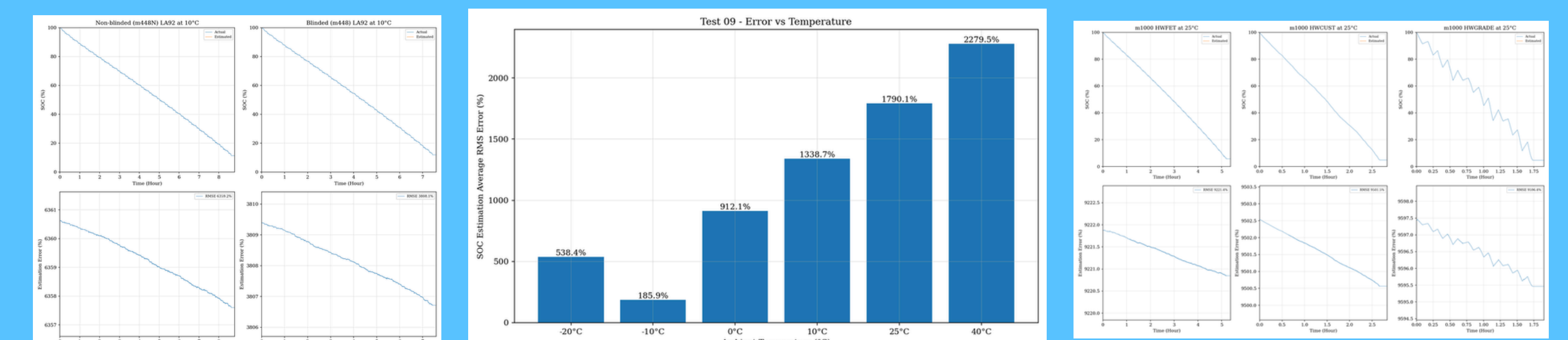
The *Dask Library* in Python with Coiled and GCP reduces CPU utilization on the host server by upwards of **90%** for heavy workloads, and reduces host's CPU usage time by **74%**

Results on 4 workers at 8GB RAM; host at 32GB RAM

Evaluator Tool

- Evaluation tool used to test and evaluate the performance of user submitted Battery SOC Estimation models
- Accepts both Python and MATLAB file types for model submission
- Python evaluator was adapted from the legacy MATLAB evaluation tool, and rewritten to fit a new-and-improved system architecture
- The evaluation tool runs a user's submitted model against several rigorous battery tests and returns its metrics and insightful graphs back to the user

Some examples of SOC graphs from the evaluation tool:



Tech Stack

